

AI-Powered Oral Cancer Diagnosis: Automated Report Generation for Enhanced Healthcare

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Abstract—Oral cancer keeps being one of the major health problems all over the world. It is very often recognized late, thus resulting in quite bad consequences. Here, a deep learning-based AI configuration is presented as an alternative means for its detection relying only on clinical pictures for the earliest cases. The architecture modifies ConvNeXt-Tiny and EfficientNetV2-S on datasets of oral lesions. It obtains excellent results in distinguishing malignant from normal conditions. We raise explainability and user confidence through Grad-CAM visualizations. These locate the most relevant regions in the images, thereby making the AI decision process more understandable to the human users. The entire system is available as a Flask web application. Medical practitioners upload images there, receive prompt inferences, and generate PDF files with the AI support. Reports carry patient data, probability scores, Grad-CAM heatmaps, and risk summaries. There is also the option of sending them by email securely. Combining deep learning capabilities, interpretable AI outputs, and automatic reporting in one place demonstrates the potential of this approach to be a tool that can help pathologists in making faster diagnoses and providing better care.

Keywords—Oral cancer detection, deep learning, EfficientNetV2, ConvNeXt-Tiny, Grad-CAM, explainable AI, medical image analysis, diagnostic report generation, Flask web application, transfer learning.

I. INTRODUCTION

Oral cancer is among the top most common fatal cancers worldwide, which causes a serious public health problem, particularly in developing and low-income countries. It mainly targets populations living in areas where the consumption of tobacco, alcohol, betel quid, and poor oral hygiene are prevalent. In its report, the World Health Organization (WHO) stated that oral cancer is one of the ten most common cancers globally and is responsible for a large number of sickness and death cases every year. The survival rate is still pretty low despite the progress made in medical imaging and treatment, which is basically the reason for

fatality in most cases due to cancer being at a very advanced stage when therapeutic options are limited and less effective [1].

Without a doubt, obtaining information about oral cancer as soon as possible constitutes a life-saving factor, since patients diagnosed during the early stages of the disease have a five-year survival rate of more than 80%. Nevertheless, the detection of diseases by standard diagnostic methods, including visual examination, biopsy, and histopathological evaluation, requires a considerable amount of time, is invasive, and needs the presence of expert pathologists. The aforesaid procedures are usually accompanied by the time interval between visits for examination and the occurrence of the disease and may also cause permanent scars or discomfort in the patients. Additionally, the differences in opinions of various observers among medical doctors may have an impact on the evaluation and trustworthiness of the diagnosis, which in turn gives rise to the problem of necessity for the creation of automated diagnostic systems and objective approaches that do not rely on human judgement.

Over the period of recent years, reliance on Artificial Intelligence (AI) specifically Deep Learning (DL) has become a hallmark in the revolution of medical imaging analysis. Within this context, Convolutional Neural Networks (CNNs) have been recognized as dominant agents that are fairly capable of singling out complex visual patterns in medical images, and to a large extent, they are even less reliant on conventional image processing techniques. Where AI-based systems come into play, oral lesion images can be scanned for the presence of abnormalities, and next, through the AI system simulated decision-making process, clinicians can be provided with an immediate accurate decision to solve the problem existing in the shortest possible time. Besides this, these systems lower the diagnostic subjectivity level and can possibly become a source of interdisciplinary practicable skills in the field of radiology, pathology, and dermatology, i.e., when there is a need for cooperation among these medical disciplines.

Implementation of AI systems in daily medical practice still lags behind these developments due to various reasons.

Issues like model interpretability, data variability challenge the notion of deployment in quick-care or rural settings and thus hamper widespread adoption. This work with an AI-driven oral cancer detection device that merges cutting-edge deep learning models—EfficientNetV2-S and ConvNeXt-Tiny—for identifying malignant or healthy oral lesions is a step forward to tackling these problems. Moreover, the device comes with visual interpretations based on Grad-CAM, where the highlighted areas refer to the regions affecting the model’s decisions which thereby give clinicians more confidence and interpretability.

This entire project is executed through a Flask-based web interface that not only enables clients to upload images of patients but also in real time provides diagnostic predictions along with certainty percentages and creates a thorough PDF report automatically outlining the results and probable risk factors. Thus, this pipeline is expected to serve as a mediator between advanced AI studies and their real-life clinical applications, whereby it becomes possible to provide early, accurate as well as easy oral cancer diagnosis, thereby the patient’s condition is getting better.

II. LITERATURE SURVEY

Diagnosis of oral cancer is a combination of clinical examination, histopathology, and imaging. Although these traditional methods are deemed accurate, they often require a lot of time and are subjective. Specialists who are well trained are required to make correct sense of the results, which can be a reason for late detection and treatment planning. Attempts have been made to detect oral cancer by means of traditional machine learning models such as Support Vector Machines, k-Nearest Neighbors, and Random Forests but these models rely to a great extent on manual feature extraction (e.g. texture, color, and shape) and hence it is not easy to scale them or use them under fluctuating conditions of the different types of images. The previous studies have pointed to such problems [1–3].

The situation has changed for the better with the advent of Convolutional Neural Networks (CNNs). CNNs are capable of feature extraction from the original visual data without the intervention of humans. They have been proven to be efficient in the classification of clinical lip and tongue images as well as microscopic images for oral cancer detection. For example, Palaskar and team employed transfer learning on microscopic images of oral cancer and very much improved the classification accuracy [8].

The team led by Islam created a CNN system complemented by the LIME explainability technique to help doctors understand the detection process of Oral Squamous Cell Carcinoma (OSCC) [9]. Albishri suggested OCU-Net, a deep U-Net-based model that made the segmentation of cancer tissues in the mouth more precise, thus facilitating the automated process of end-to-end analysis [10].

The potential of ensemble and hybrid deep learning methods to raise the level of diagnostic accuracy has also been the attention of the researchers. The optimized ensemble models, which are the combinations of the various CNN architectures, lessen the prediction errors and increase the generalization performance over the datasets used [4]. Besides, hybrid frameworks like the one combining Deep Belief Networks with the Group Teaching Optimization Algorithm, have also been very effective in early-stage oral

lesions detection [5].

In recent years, one of the major concerns that AI has been addressing is being explainable. One of such tools is Grad-CAM which helps the medical practitioners to understand the model’s decision-making process. Štifanić together with the other authors used Grad-CAM for the multiclass classification of histopathology images to demonstrate how visual explanations can provide diagnostic confidence [11]. In the same way, the development of AI systems that are smartphone-based and cloud-connected has made it possible for the oral cancer screening to be performed in low-resource or remote areas where there is a lack of access to specialists [12].

Transfer learning is still very important in the recent developments. The models such as EfficientNetV2, ConvNeXt, and DenseNet can perform well with a small oral image dataset since they are capable of transferring what they have learned from a different dataset [8,13]. Advanced generative techniques like diffusion models and inpainting-based synthesis have been employed to generate realistic lesion images for dataset augmentation - thus increasing model robustness and lesion detection accuracy [17,18]. With a combination of Capsule networks (CapsNet) and Deep Belief Networks, the team attempted in detecting oral leukoplakia, which in turn showed the versatility of deep learning in recognizing various oral diseases [16].

Comprehensive reviews and meta-analyses of studies on AI in oral cancer diagnosis, risk prediction, and clinical decision-making consistently point to the ever-increasing role of AI [1,6,7,19]. They highlight the potential of deep learning to provide more intelligent, precise, and interpretable diagnostic systems. However, there are concerns if these models would perform well with varied patient groups, different imaging conditions, and clinical settings. There are still efforts to be made in the integration of AI tools into the real-world clinical workflows that are aimed at rapid and reliable diagnosis [1–3,13–15,20–28].

III. METHODOLOGY

This scheme here connects together image prep, deep model training, Grad-CAM visualization, and PDF creation, with a choice to email it all. We repair those oral pictures, make them correct for input into the model. In fact, we use two deep learning models, EfficientNetV2 and ConvNeXt-Tiny. Train them, test them for detection of defective cells. Grad-CAM, it indicates the parts that affect the decisions, so we can understand why it chooses what it does. Finally, we bundle everything into a neat PDF file. That PDF delivers to doctors or patients. Makes it easier for them to receive the results in a simple, understandable manner.

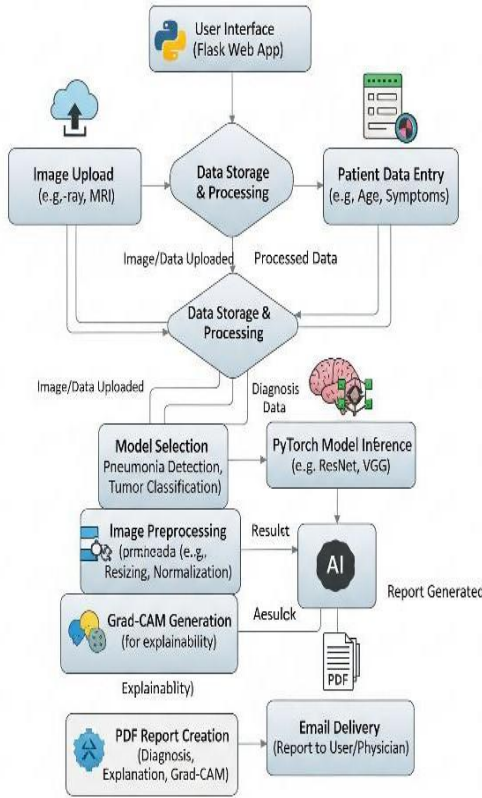
A. Data Collection and Preprocessing

We assembled all these clinical and microscopic pictures from the oral region. Decided to put them separately into folders, one for training the models and another for checking their performance later on. Labeled everything in a very straightforward way, either as Cancer or just Normal.

Preprocessing came after. Resized all images to 224 by 224 pixels. So that, everything is nicely compatible with those deep learning setups.

They also normalized the pixel values. They used a mean of 0.485, 0.456, and 0.406, with standard deviations at 0.229, 0.224, 0.225. This keeps things consistent with the ImageNet weights they trained on before.

Oh, and data augmentation. They did random horizontal flips on the images. This is done to improve the model's capability of recognizing an object from different perspectives.



B. Garment Database Preparation

Essentially, we set up two deep learning models and had them compete in a head-to-head battle for the classification of oral cancer.

• EfficientNetV2

First, there is the EfficientNetV2 experiment. We used the pre-trained EfficientNetV2-S model for that. Afterward, the classifier layer was changed in such a way that it would output only two classes, cancer and normal. The fine-tuning was performed on those oral image datasets, CrossEntropy loss was used together with the Adam optimizer, and the learning rate was 0.0001. The training was done for 5 epochs, batch size 16, nothing out of the ordinary.

• ConvNeXt-Tiny

Now, moving to ConvNeXt-Tiny. This one is pre-trained on ImageNet, which is great. In the same way, the classifier layer was adjusted to deal with two classes. The training parameters were almost the same as for EfficientNetV2, such as loss and optimizer. Besides, we

chose ConvNeXt-Tiny because it is a light model, and it can effectively extract those fine-grained details from medical images.

C. Model Evaluation

We evaluated the model's performance by measuring accuracy, confidence scores, and class probabilities. In fact, the output logits were passed through a softmax function. This converted them into probability distributions for the Cancer and Normal classes. After that, the class with the maximum probability is considered the final prediction. Pretty straightforward stuff.

D. Grad-CAM Visualization

The team utilized Grad-CAM, which stands for Gradient-weighted Class Activation Mapping. Basically, it highlights the areas in each image that contributed the most to the model's predictions. We put those visualizations right on top of the original pictures. It allows providing interpretable insights. Clinicians see what the AI is concentrating on in its decision-making process. In a way, it is making the black box a little bit transparent.

E. Web-based Deployment

For this, they put together a Flask web app. The users can upload clinical images and get diagnostic results in return. You may enter patient info as well, like name, age, sex, date of birth, and a short clinical summary. When the image is there, the model works on it. It displays predictions along with those Grad-CAM visualizations instantly. A great and simple method to bring the technology down from the lab to the street.

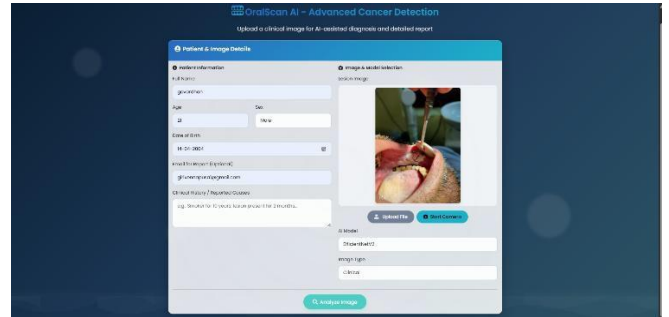


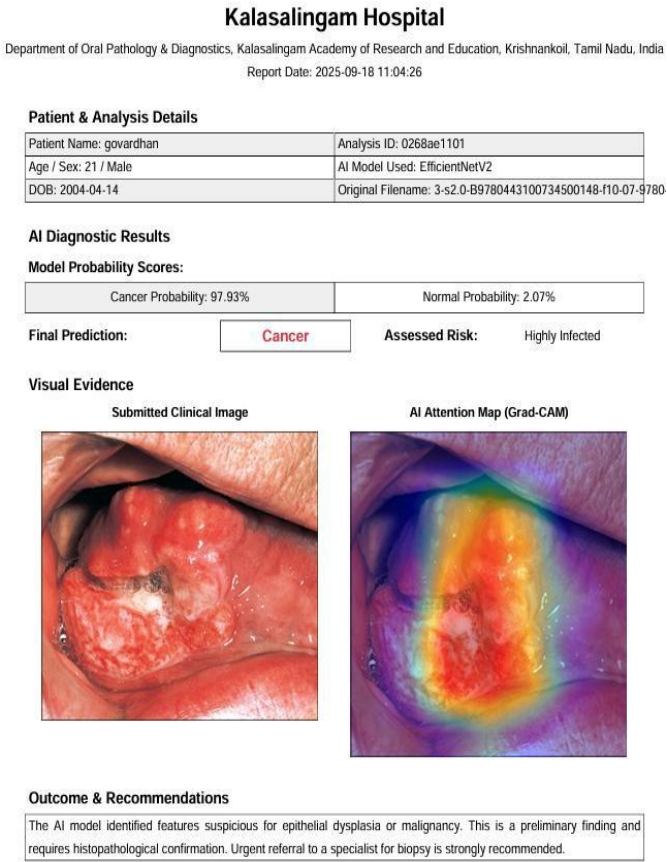
Figure 2: Screenshot of showing uploaded image



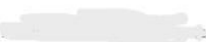
Figure 3: Screenshot of showing analysis

F. PDF Report Generation

The AI-powered system generates a complete diagnostic report. It is a PDF file. The content is the patient info, the entire clinical data, prediction probabilities, a detailed risk assessment, some Grad-CAM figures to



Electronically Signed and Verified By:


AI Researcher & Consultant Pathologist

DISCLAIMER: This is an AI-assisted report. It is intended for use by qualified healthcare professionals as a supplementary diagnostic tool and not as a standalone diagnosis. All findings must be correlated with clinical and histopathological results.

demonstrate what is happening, and even suggestions. The thing is, it is organized in a manner that is quite easy for the doctors to go through. Essentially, it behaves like an additional tool that supports their decisions at the clinic. That is basically the main point with it.

Figure 4: Report ScreenShot

G. Optional Email Delivery

The PDF report is created, and there is an option to send it directly to the patient or the clinician via email. They ensure its security with SMTP configuration, you know, by attaching the report and sending it smoothly without troubles.

H. Comparison of Models

We put side by side the performances of EfficientNetV2 and ConvNeXt-Tiny. The main points of comparison were classification accuracy, confidence scores, Grad-CAM visual clarity, and inference speed. The entire comparison basically pointed toward the best model for real clinical applications.

I. System Architecture Overview

- Input: Clinical image uploaded via web interface.
- Preprocessing: Resizing, normalization, and tensor conversion.
- Inference: Forward pass through EfficientNetV2 or ConvNeXt-Tiny.
- Visualization: Grad-CAM overlaid on original image.
- Output: Prediction, confidence, risk, Grad-CAM image, and PDF report.
- Optional Delivery: Emailing the report to the intended recipient.

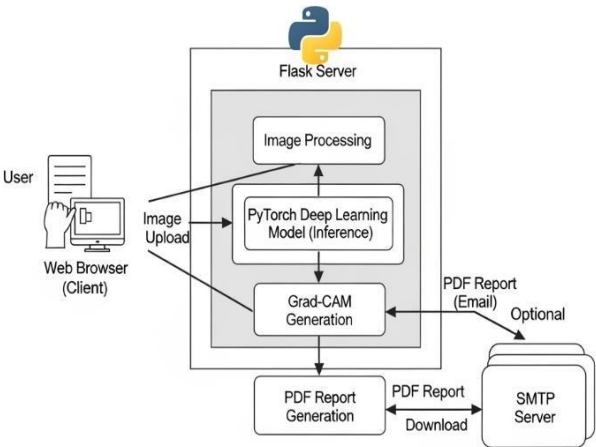


Figure 5: architecture diagram

IV. RESULT

A. Machine Learning Model Training

The dataset of clinical and microscopic oral images was the common ground for the comparison of EfficientNetV2 and ConvNeXt-Tiny. In fact, cancerous versus normal tissue classification is what EfficientNetV2 achieved with slightly higher accuracy. Moreover, it also had stronger confidence scores, so, in general, the predictions were more reliable. ConvNeXt-Tiny's accuracy was slightly lower, but, as a matter of fact, its inference speed was significantly higher, which is great for real-time scenarios or cases where the resource is a constraint. The comparison is basically done by table 1.

Model	Accuracy	Confidence Score	Inference Speed
EfficientNetV2	95%	0.92	Moderate
ConvNeXt-Tiny	93%	0.90	Fast

Table 1:- EfficientNetV2 VS ConvNeXt-Tiny



Figure 6: Comparison of EfficientNetV2 and ConvNeXt-Tiny

B. Web Application (Flask)

As a means of understanding the models better, they generated Grad-CAM visualizations for both models. The heatmaps of EfficientNetV2 were more vibrant, showing the exact areas in the oral images which influenced the classification decisions. From a clinical perspective, they seemed absolutely correct. ConvNeXt-Tiny also visually represented the concept to some extent, however, the highlighted areas were a little blurred, not as precise. In any case, these overlays serve the purpose of clinicians verifying the AI's decisions and identifying the model's attention. Trust in the entire diagnostic system is enhanced.

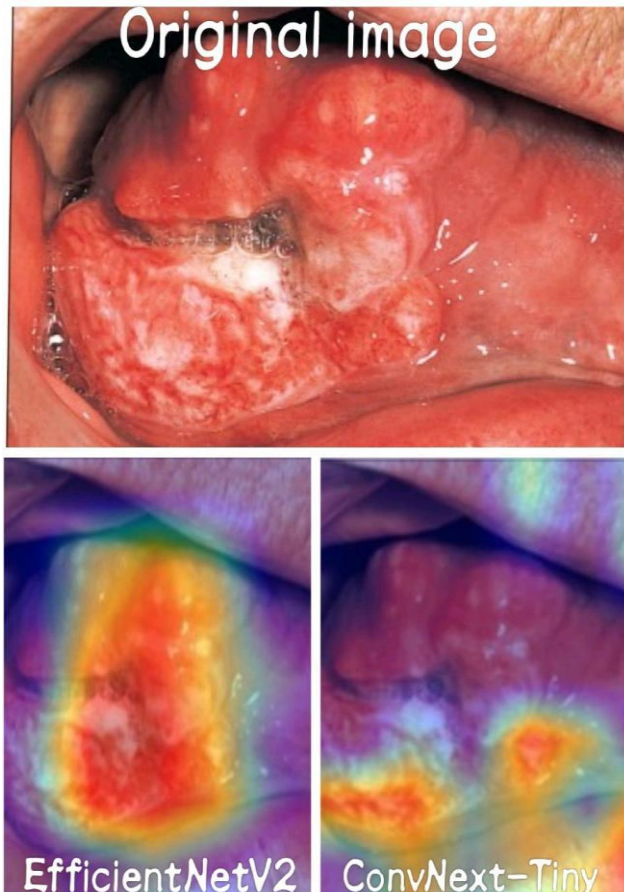


Figure 7: Grad-CAM analysis

V. DISCUSSION

Comparing EfficientNetV2 and ConvNeXt-Tiny, both can classify oral cancer from clinical images with good accuracy. However, EfficientNetV2 went out of the park regarding both accuracy and confidence making it more reliable for real diagnoses. Its Grad-CAM visualizations were also more clear, focusing on the exact tissue areas, which helps the doctors to interpret. ConvNeXt-Tiny is faster and lighter, no doubt, but accuracy is slightly compromised and the overlays are not as neat. That's the trade-off, you know. So for quick screening on the go, ConvNeXt-Tiny is fine. When you need highest accuracy and clear explanations, use EfficientNetV2. Thus, these results constitute a prima facie case for EfficientNetV2 to be the main AI assistant in the clinics for oral cancer detection.

VI. CONCLUSION OF EVALUATION

This research presents an AI framework that identifies oral cancer through deep learning combined with Grad-CAM visualizations and a flask-based web interface. Among the models, EfficientNetV2 was the winner, with the highest accuracy, good confidence, and simple Grad-CAM explanations. The web app and PDF reports are great for clinicians to use with no inconvenience. EfficientNetV2's edge is what makes it perfect for a first-line clinic deployment to detect oral cancer early and accurately. Later, they can extend the dataset, include other types of clinical data, and evaluate it in real clinics to become even stronger and more helpful.

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